

Joint Inspection Report

Date:- 06.08.2026

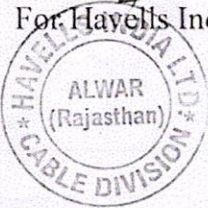
Minutes of meeting held between M/s Havells India Ltd-Alwar & APDCL regarding inspection of 33 KV MVCC, 1CX241 SQ.MM.

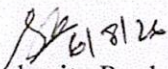
Customer:- M/S UK ENTERPRISE.

Insp. Memo No:- MD/APDCL/HR/Misc./2025/36/42 Dt. 01.08.2026

- 1) M/s Havells India Limited offered the cables for Inspection as per enclosed offer/Packing List.
- 2) Inspection was carried out on 04.08.2026 to 06.08.2026 at the Works of M/s Havells India Limited, A-461/462 & SP-215, MIA, ALWAR (Rajasthan).
- 3) All acceptance tests were witnessed on randomly selected cable drums as per relevant spec. EN 50397-1 with up to date amendments.
- 4) All the test results are found within permissible limits. The material are technically acceptable and all the offered lot as per attached packing list are cleared for dispatch at manufacturer works.
- 5) Internal test reports reviewed.
- 6) The following documents were attached by M/s Havells India Limited:
 - Offer/Packing List
 - Acceptance Test Report
 - Internal test reports
 - Calibration status of testing equipment's


For Havells India Ltd.




Sushmita Paul
Dy. Manager, O/o CGM(PP&D)-APDCL



HAVELLS INDIA LIMITED

A-461/462, M.I.A., Alwar (Rajasthan)

OFFER / PACKING LIST

DATE:- 04.08.2026 to 06.08.2026

Customer:-

M/S UK ENTERPRISE.

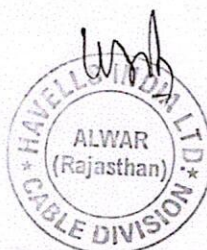
Spec ref:-

EN:50397-1(In Genl.), IS:398 PT-4 WITH UPTO DATE AMENDMENTS

OA NO:-

116608787/Y

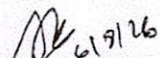
S.No.	Description of material	Drum No.			Lengths (mtrs)		
1.1	1CX241 SQ.MM AAAC ANTI TRACK 33 KV COVERED CONDUCTOR	HT	17516	/26	1000	-	1000
2		HT	17517	/26	1007	-	1007
3		HT	17526	/26	1000	-6*	994
4		HT	17527	/26	1000	-	1000
5		HT	17528	/26	1007	-	1007
6		HT	17529	/26	1007	-	1007
7		HT	17530	/26	1000	-	1000
8		HT	17531	/26	1000	-	1000
9		HT	17532	/26	1000	-	1000
10		HT	17533	/26	1005	-	1005
11		HT	17540	/26	1000	-	1000
12		HT	17541	/26	1010	-	1010
13		HT	17542	/26	1015	-	1015
14		HT	17543	/26	995	-	995
15		HT	17544	/26	995	-6*	989
16		HT	17545	/26	1013	-	1013
17		HT	17546	/26	1010	-	1010
18		HT	17571	/26	1022	-	1022
19		HT	17574	/26	1000	-	1000
20		HT	17575	/26	1000	-	1000
21		HT	17576	/26	1001	-	1001
22		HT	17577	/26	990	-	990
23		HT	17578	/26	990	-	990
24		HT	17585	/26	1028	-	1028
25		HT	17586	/26	1025	-	1025
26		HT	17587	/26	1000	-	1000
27		HT	17588	/26	995	-	995
28		HT	17589	/26	995	-6*	989
29		HT	17590	/26	1000	-	1000
30		HT	17591	/26	1016	-	1016
31		HT	17592	/26	1010	-	1010
32		HT	17593	/26	1008	-	1008
33		HT	17594	/26	972	-	972
34		HT	17595	/26	1010	-	1010
35		HT	17596	/26	1011	-	1011



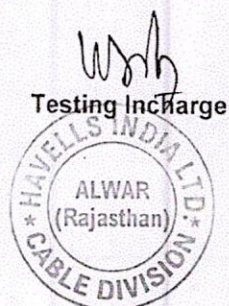
36		HT	17599	/26	1000	-	1000
37		HT	17602	/26	990	-	990
38		HT	17607	/26	678	-6*	672
39		HT	17612	/26	1000	-	1000
40		HT	17613	/26	1000	-	1000
41		HT	17614	/26	1170	-	1170
42		HT	17615	/26	1009	-	1009
43		HT	17620	/26	989	-	989
44		HT	17621	/26	1183	-	1183
45		HT	17623	/26	980	-	980
46		HT	17624	/26	999	-	999
47		HT	17625	/26	980	-	980
48		HT	17627	/26	996	-6*	990
49		HT	17628	/26	996	-	996
50		HT	17629	/26	981	-	981
51		HT	17630	/26	980	-	980
52		HT	17633	/26	990	-	990
53		HT	17634	/26	996	-	996
54		HT	17636	/26	1000	-6*	994
55		HT	17637	/26	999	-	999
56		HT	17638	/26	1001	-	1001
57		HT	17639	/26	950	-	950
58		HT	17640	/26	904	-	904
59		HT	17641	/26	1000	-	1000
		TOTAL----->			58908	-36*	58872

*Marked lengths consumed in testing during inspection.

Above material inspected ok & cleared for dispatch.


Sushmita Paul

Dy. Manager, O/o CGM(PP&D)-APDCL





HAVELLS

HAVELLS INDIA LIMITED
A-461/462, M.I.A., Alwar (Rajasthan)
ACCEPTANCE(SAMPLE) TEST RESULTS

DATE:- 04.08.2026 to 06.08.2026

Customer

M/S UK ENTERPRISE.

Description of cable:-

1CX241 SQ.MM AAC ANTI TRACK 33 KV COVERED CONDUCTOR

Ref spec:-

EN:50397-1(In Genl.), IS:398 PT-4 WITH UPTO DATE AMENDMENTS

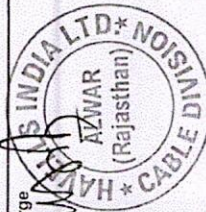
TEST DESCRIPTION	UNITS	Specified Values	Observed Values	Observed Values	Observed Values	Observed Values
I) Test on conductor			D. No: HT17526/26	D. No: HT17544/26	D. No: HT17589/26	D. No: HT17607/26
a) Details of conductor	-	-	Stranded Circular AAAC conductor with water blocking material.			
b) Breaking load(RTS) of conductor(Min.)	KN	70.61	76.15	78.56	79.56	74.56
c) Conductor Resistance (Max.) at 20 Deg C	Ohm/KM	0.143	0.126	0.125	0.128	0.127
d) No. of strands(Min.)	nos.	19	19	19	19	19
e) Diameter of individual strand(approx.)	mm	4.02	4.03	4.05	4.04	4.03
II) Electrical Tests						
a) High voltage test @ 33 kV AC for 4 hours on sample	Should Withstand Without Any Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown	No Breakdown
b) Insulation Resistance/Volume Resistivity						
~ At 27°C	ohm-cm	Min. 1X10 ¹⁴	2.5 X 10 ¹⁵	5.2 X 10 ¹⁵	3.8 X 10 ¹⁵	5.6 X 10 ¹⁵
~ At 90°C	ohm-cm	Min. 1X10 ¹²	1.6 X 10 ¹⁵	4.3 X 10 ¹⁵	2.7 X 10 ¹⁵	3.2 X 10 ¹⁵
III) Physical Dimensions						
a) Overall diameter(approx.)	mm	30	30.15	29.86	30.15	30.58
b) Thickness of outer sheath covering(Nom.)	mm	1.20	1.32	1.39	1.33	1.36
c) Thickness of insulation(Nom.)	mm	2.43	2.55	2.53	2.56	2.59
d) Thickness of conductor screening(approx.)	mm	0.30	0.54	0.51	0.53	0.52
IV) Physical tests						
a) Tensile strength	N/mm ²	Min. 12.5	19.56	17.15	18.05	21.15
b) Elongation at break	%	Min. 200	500	495	510	520
c) Hot set test	%					
~ Elongation under load	%	Max. 175	92	85	74	75
~ Permanent Set	%	Max. 15	2.65	2.05	1.45	1.50
V) Shrinkage test (Max.)	%	4	2.00	1.25	0.75	1.50
VI) Lay Ratio	-	10-14	11.56	12.06	11.69	12.03
VII) Dimension/verification of wooden drums:- All offered drums verified and found as per drawing/specification requirement.						
VIII) Checks for length, surface finish	All samples drum out of offered lot are verified and found in order.					
IX) Marking of drums:- All offered drums verified and found as per drawing/specification requirement.						
Tested By						
Test Witnessed by						

Testing Incharge

Sushmita Paul

06/08/26
Sushmita Paul

Dy. Manager, O/o CGM(PP&D)-APDCL





HAVELLS

HAVELLS INDIA LIMITED
A-461/462, M.I.A., Alwar (Rajasthan)
ACCEPTANCE(SAMPLE) TEST RESULTS

DATE:- 04.08.2026 to 06.08.2026

Customer

M/S UK ENTERPRISE.

Description of cable:-

1CX241 SQ.MM AAAC ANTI TRACK 33 KV COVERED CONDUCTOR

Ref spec:-

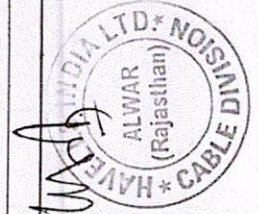
EN:50397-1(In Genl.), IS:398 PT-4 WITH UPTO DATE AMENDMENTS

TEST DESCRIPTION	UNITS	Specified Values	Observed Values	D. No: HT17636/26
I) Test on conductor				
a) Details of conductor	-	-	Stranded Circular AAAC conductor with water blocking material.	
b) Breaking load(RTS) of conductor(Min.)	KN	70.61	76.45	
c) Conductor Resistance (Max.) at 20 Deg C	Ohm/KM	0.143	0.127	
d) No. of strands(Min.)	nos.	19	19	
e) Diameter of individual strand(approx.)	mm	4.02	4.05	
II) Electrical Tests				
a) High voltage test @ 33 KV AC for 4 hours on sample	Should Withstand Without Any Breakdown	No Breakdown		
b) Insulation Resistance/Volume Resistivity				
~ At 27°C	ohm-cm	Min. 1×10^{14}	4.2×10^{15}	
~ At 90°C	ohm-cm	Min. 1×10^{12}	3.6×10^{15}	
III) Physical Dimensions				
a) Overall diameter(approx.)	mm	30	30.59	
b) Thickness of outer sheath covering(Nom.)	mm	1.20	1.36	
c) Thickness of insulation(Nom.)	mm	2.43	2.55	
d) Thickness of conductor screening(approx.)	mm	0.30	0.52	
IV) Physical tests				
a) Tensile strength				
~ INSULATION	N/mm ²	Min. 12.5	18.15	
b) Elongation at break				
~ INSULATION	%	Min. 200	535	
c) Hot set test				
~ Elongation under load	%	Max. 175	75	
~ Permanent Set	%	Max. 15	2.70	
V) Shrinkage test (Max.)	%	4	1.25	
VI) Lay Ratio	-	10-14	11.95	
VII) Dimension/Verification of wooden drums:- All offered drums verified and found as per drawing/specification requirement.				
VIII) Checks for length, surface finish	All samples drum out of offered lot are verified and found in order.			
IX) Marking of drums:- All offered drums verified and found as per drawing/specification requirement.				

Tested By

Test Witnessed by

Testing Incharge



[Signature]
Sushmita Paul

Dy. Manager, O/o CGM(PP&D), APDCL



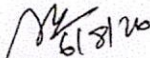
HAVELLS INDIA LIMITED
A-461/462, M.I.A., Alwar (Rajasthan)
TEST REPORT

DATE:- 04.08.2026 to 06.08.2026

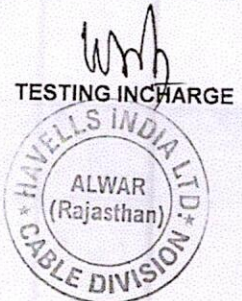
Customer:- M/S UK ENTERPRISE.
Spec ref:- EN:50397-1(In Genl.), IS:398 PT-4 WITH UPTO DATE AMENDMENTS
Size:- 1CX241 SQ.MM AAAC ANTI TRACK 33 KV COVERED CONDUCTOR
DRUM NO. HT17526/26

Sr. No.	TESTS	SPECIFIED LIMIT	OBTAINED VALUE	REMARK
1	Tracking Resistance Annex C of EN 50397 -1 : 2006	Length of specimen= 200 mm, Test voltage= 3.5 kV Contaminant flow rate 0.30 ml/min Series resistance = 22 k ohm Contaminant = Ammonium chloride Specimen mounting angle= 45 Test duration = 6 Hours The test is successful if the current in the high voltage circuit does not exceed 60 mA during 6 hours	Withstood	Conforms
2	Leakage Current , mA	Max. 1.0	0.061	Conforms
3	Longitudinal water penetration test : sample cut which has been subjected to bending test, a ring approx. 50 mm width is cut in the middle of the sample and all covering over the moisture barrier shall be removed.	No traces of water should observe at the ends of sample during test folowed by bending test	No traces of water observed (Passes the test)	Conforms

Test witnessed By


Sushmita Paul

Dy. Manager, O/o CGM(PP&D)-APDCL





HAVELLS INDIA LTD.

A-461/462, M.I.A., Alwar (Rajasthan)

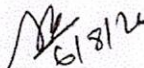
REWINDING STATEMENT

DATE:- 04.08.2026 to 06.08.2026

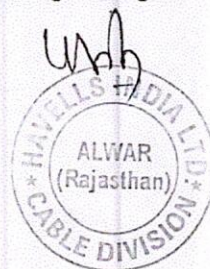
S.No.	SIZE	DRUM No.	Offer Lengths (Mtrs.)	Observed length (mtrs.)	Surface finish, Visual checks
1	1CX241 SQ.MM AAAC ANTI TRACK 33 KV COVERED CONDUCTOR	HT17588/26	995	995	OK

Witnessed By

Testing incharge


Sushmita Paul

Dy. Manager, O/o CGM(PP&D)-APDCL





HAVELLS INDIA LIMITED (CABLE DIVISION)

Works : A 461/462 M.I.A. , Alwar (Raj)-301030

ROUTINE TEST CERTIFICATES

DATE:- 23.07.2026

Customer

M/S UK ENTERPRISE.

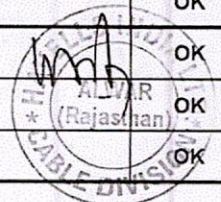
Ref. Spec

EN:50397-1(In Genl.), IS:398 PT-4 WITH UPTO DATE AMENDMENTS

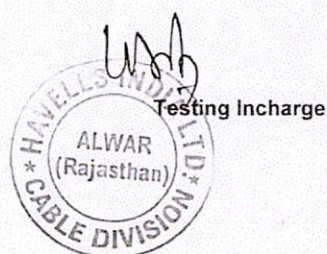
Description of Cable

1CX241 SQ.MM AAAC ANTI TRACK 33 KV COVERED CONDUCTOR

Drum no			Length (Mtrs.)	High Voltage Test at 33 kV AC for 4 Hours	Spark Test 0.7 U (AC)	Conductor Resistance at 20°C in Ohms/km	Remark
HT	17516	/26	1000	No Breakdown	No Breakdown	0.126	OK
HT	17517	/26	1007	No Breakdown	No Breakdown	0.128	OK
HT	17526	/26	1000	No Breakdown	No Breakdown	0.125	OK
HT	17527	/26	1000	No Breakdown	No Breakdown	0.124	OK
HT	17528	/26	1007	No Breakdown	No Breakdown	0.125	OK
HT	17529	/26	1007	No Breakdown	No Breakdown	0.127	OK
HT	17530	/26	1000	No Breakdown	No Breakdown	0.125	OK
HT	17531	/26	1000	No Breakdown	No Breakdown	0.126	OK
HT	17532	/26	1000	No Breakdown	No Breakdown	0.124	OK
HT	17533	/26	1005	No Breakdown	No Breakdown	0.125	OK
HT	17540	/26	1000	No Breakdown	No Breakdown	0.126	OK
HT	17541	/26	1010	No Breakdown	No Breakdown	0.125	OK
HT	17542	/26	1015	No Breakdown	No Breakdown	0.124	OK
HT	17543	/26	995	No Breakdown	No Breakdown	0.127	OK
HT	17544	/26	995	No Breakdown	No Breakdown	0.127	OK
HT	17545	/26	1013	No Breakdown	No Breakdown	0.125	OK
HT	17546	/26	1010	No Breakdown	No Breakdown	0.124	OK
HT	17571	/26	1022	No Breakdown	No Breakdown	0.126	OK
HT	17574	/26	1000	No Breakdown	No Breakdown	0.125	OK
HT	17575	/26	1000	No Breakdown	No Breakdown	0.125	OK
HT	17576	/26	1001	No Breakdown	No Breakdown	0.126	OK
HT	17577	/26	990	No Breakdown	No Breakdown	0.124	OK
HT	17578	/26	990	No Breakdown	No Breakdown	0.125	OK
HT	17585	/26	1028	No Breakdown	No Breakdown	0.126	OK
HT	17586	/26	1025	No Breakdown	No Breakdown	0.125	OK
HT	17587	/26	1000	No Breakdown	No Breakdown	0.126	OK
HT	17588	/26	995	No Breakdown	No Breakdown	0.124	OK
HT	17589	/26	995	No Breakdown	No Breakdown	0.125	OK
HT	17590	/26	1000	No Breakdown	No Breakdown	0.126	OK
HT	17591	/26	1016	No Breakdown	No Breakdown	0.125	OK



HT	17592	/26	1010	No Breakdown	No Breakdown	0.124	OK
HT	17593	/26	1008	No Breakdown	No Breakdown	0.125	OK
HT	17594	/26	972	No Breakdown	No Breakdown	0.126	OK
HT	17595	/26	1010	No Breakdown	No Breakdown	0.124	OK
HT	17596	/26	1011	No Breakdown	No Breakdown	0.126	OK
HT	17599	/26	1000	No Breakdown	No Breakdown	0.125	OK
HT	17602	/26	990	No Breakdown	No Breakdown	0.124	OK
HT	17607	/26	678	No Breakdown	No Breakdown	0.127	OK
HT	17612	/26	1000	No Breakdown	No Breakdown	0.126	OK
HT	17613	/26	1000	No Breakdown	No Breakdown	0.125	OK
HT	17614	/26	1170	No Breakdown	No Breakdown	0.127	OK
HT	17615	/26	1009	No Breakdown	No Breakdown	0.125	OK
HT	17620	/26	989	No Breakdown	No Breakdown	0.126	OK
HT	17621	/26	1183	No Breakdown	No Breakdown	0.126	OK
HT	17623	/26	980	No Breakdown	No Breakdown	0.125	OK
HT	17624	/26	999	No Breakdown	No Breakdown	0.124	OK
HT	17625	/26	980	No Breakdown	No Breakdown	0.126	OK
HT	17627	/26	996	No Breakdown	No Breakdown	0.127	OK
HT	17628	/26	996	No Breakdown	No Breakdown	0.126	OK
HT	17629	/26	981	No Breakdown	No Breakdown	0.126	OK
HT	17630	/26	980	No Breakdown	No Breakdown	0.127	OK
HT	17633	/26	990	No Breakdown	No Breakdown	0.125	OK
HT	17634	/26	996	No Breakdown	No Breakdown	0.126	OK
HT	17636	/26	1000	No Breakdown	No Breakdown	0.127	OK
HT	17637	/26	999	No Breakdown	No Breakdown	0.126	OK
HT	17638	/26	1001	No Breakdown	No Breakdown	0.125	OK
HT	17639	/26	950	No Breakdown	No Breakdown	0.128	OK
HT	17640	/26	904	No Breakdown	No Breakdown	0.126	OK
HT	17641	/26	1000	No Breakdown	No Breakdown	0.125	OK
SPECIFIED VALUES ---->				Should Withstand Without Any Breakdown	Should Withstand Without Any Breakdown	Max.	
						0.143	



CALIBRATION STATUS OF TESTING EQUIPMENTS

Sr. No.	Equipment Name	Make/Model	Sr./ID No. of equipment	Range & Least Count	Calibration by laboratory	Date of Calibration	Next Date of Calibration	Lab Certificate No.
1	Million Mega Ohm Meter	Shivananda Electronics/ LS-3B	PALTLRM009	10-1000X10 ⁶ MMQ/500V DC	Excellent Instrument Calibration Private Limited	11-04-2026	10-04-2027	CC-2597
2	Resistance Meter (Insulation Tester)	Johri	PALTLRM001	0 - 19.99KQ	Excellent Instrument Calibration Private Limited	25-11-2025	24-11-2026	CC-2597
3	Digital Tensile Testing Machine	Presto	1407154763 (PALTLTE001)	0 to 500N/0.1N	Excellent Instrument Calibration Private Limited	26-11-2025	25-11-2026	CC-2597
4	Digital Tensile Testing Machine (Heavy Duty)	Presto	3108154999 (PALTLTE004)	0 to 1000kgf/1kgf	Excellent Instrument Calibration Private Limited	26-11-2025	25-11-2026	CC-2597
5	Profile Projector	Escale	PALTLPP001	0-100mm / 0.005mm	Excellent Instrument Calibration Private Limited	08-07-2026	07-07-2027	CC-2597
6	Steel Scale	Omega	PALTLSC005	0 to 1000 mm / 0.5 & 1 mm	Excellent Instrument Calibration Private Limited	25-11-2025	24-11-2026	CC-2597
7	Digital Anemometer	Lutron / AM-4201	PALTLAN001	0.4 to 30m/s / 0.1m/s	Excellent Instrument Calibration Private Limited	25-11-2025	24-11-2026	CC-2597
8	Digital Vernier Caliper	Mitutoyo	A21010369/PALTLVCO11	0 to 200 mm / 0.01 mm	HAVELLS INDIA LIMITED	01-05-2026	31-10-2026	CC-3903
9	Digital Timer	Counttron/1050	PALTLTM005	0 to 99.99 min / 0.01 min	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
10	Digital Temperature Controller with RTD Sensor (Hot Set - 0977)	VERTEX/VT4826+	PALTLTC019	0 to 400°C / 1°C	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
11	AC High Voltage Test Set	Shanghai Lanpo	PACTLHV009	0 to 125/250/350KV (L.C. 0.1 kV)	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
12	AC High Voltage Test Set	Rectifier & Electronics	PALTLHV001	0 to 50/100/200KV (L.C. 2 kV)	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
13	Conditioning Chamber	Flexem	PALTLHC007	0 to 100 % RH / 0.1 % RH	ANGEL CALIBRATION LAB LLP	03-01-2026	02-01-2027	CC-4548
14	Digital Temperature Controller with K Type Thermocouple(HCL Test)	VERTEX/VT4826+	PALTLTC033	0 to 900°C / 1°C	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
15	Partial Discharge Calibrator	Shanghai Lanpo	PACTLMF002	5pC to 100pC	SUDUTT Elec. Testing Laboratory (OPC) Pvt. Ltd.	13-07-2026	12-07-2027	CC-4335
16	PD Detector	KVTEK/D22	230/PACTLMF009	0.01-1000 pC	SUDUTT Elec. Testing Laboratory (OPC) Pvt. Ltd.	13-07-2026	12-07-2027	CC-4335
17	PD Calibrator (IN-BUILT)	KVTEK/D22	230/PACTLMF008	5 pC-1000 pC	SUDUTT Elec. Testing Laboratory (OPC) Pvt. Ltd.	13-07-2026	12-07-2027	CC-4335
18	Digital Micrometer	Mitutoyo	67576367/PALTLMI005	0 to 25 mm / 0.001 mm	HAVELLS INDIA LIMITED	09-07-2026	08-01-2027	CC-3903
19	Weighing Balance	Interface	202442/PALTLWS002	0 to 5 kg / 0.001 kg	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
20	Smoke Density	SA Associates/SDA 200	20240820/PALTLHC006	0 to 100 % / 0.1 %	MEASUREXPRT TECHCOM Systems(OPC)	18-08-2025	17-08-2026	CC-3972
21	Digital Micro Ohm Meter	RTS/RMO-36	3001/PALTLRM002	0 to 1000 milliohm	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
22	AC High Voltage Test Set	Teleflex. Electronic	PACTLHV011	0 to 5/kV/0.2KV	HAVELLS INDIA LIMITED	06-12-2025	05-12-2026	CC-3903
23	Oxygen Analyzer	SA Associates /Oxymat 3	20241020/PALTLLOS003	0 to 100% / 0.1%	MEASUREXPRT TECHCOM Systems(OPC)	18-08-2025	17-08-2026	CC-3972
24	Air Flow Meter (Fitted with Flammability Test Apparatus)	Hi-Flow	PALTLAF015	10 to 110 lpm / 5 lpm	Excellent Instrument Calibration Private Limited	20-11-2025	19-11-2026	CC-2597
25	Air Flow Meter (Fitted with Flammability Test Apparatus)	Hi-Flow	PALTLAF016	1 to 12 lpm/0.5 lpm	Excellent Instrument Calibration Private Limited	20-11-2025	19-11-2026	CC-2597

